

Tutorial 3

Matrix inverse and solving linear systems

Homework

1. Questions 2 and 6: Provide detailed solutions to *all* parts of both questions.
2. Questions 5, 7–11, 14, and 15: Select any three questions and provide detailed solutions to each selected question in its *entirety*.
3. Questions 16–18: Select any one question and provide a detailed solution to that question in its *entirety*.
4. Question 19: Provide detailed solutions to *all* parts.
5. Question 20: Provide detailed solutions to any two subparts of your choice from Parts 2–8.
6. Question 21: Provide detailed solutions to any two subparts of your choice.
7. Question 24: Provide detailed solutions to any two subparts of your choice.
8. Questions 28–30: Provide detailed solutions to any two subparts of your choice selected from Questions 28, 29, and 30. The two subparts may be selected independently across these questions.

Question 1. Confirm the validity of the given statements for the following matrices and scalars.

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 2 \\ 1 & -4 \end{pmatrix}, \quad C = \begin{pmatrix} 4 & 1 \\ -3 & -2 \end{pmatrix}, \quad \alpha = 4, \quad \beta = -7$$

- | | |
|---|---|
| 1. $(A + B) + C = A + (B + C)$ | 2. $(AB)C = A(BC)$ |
| 3. $(\alpha + \beta)C = \alpha C + \beta C$ | 4. $\alpha(BC) = (\alpha B)C = B(\alpha C)$ |
| 5. $A(B - C) = AB - AC$ | 6. $(B + C)A = BA + CA$ |
| 7. $\alpha(\beta C) = (\alpha\beta)C$ | 8. $(A^\top)^\top = A$ |
| 9. $(AB)^\top = B^\top A^\top$ | 10. $(A + B)^\top = A^\top + B^\top$ |
| 11. $(\alpha C)^\top = \alpha C^\top$ | 12. $A(B + C) = AB + AC$ |

Question 2. Find the inverse of the given matrix using the formula for computing the inverse of a 2×2 matrix.

- | | |
|--|---|
| 1. $A = \begin{pmatrix} 2 & -3 \\ 4 & 4 \end{pmatrix}$ | 2. $B = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ |
| 3. $C = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ | 4. $D = \begin{pmatrix} 6 & 4 \\ -2 & -1 \end{pmatrix}$ |

Question 3. Find the inverse of the given matrix

$$1. \begin{pmatrix} \frac{1}{2}(e^x + e^{-x}) & \frac{1}{2}(e^x - e^{-x}) \\ \frac{1}{2}(e^x - e^{-x}) & \frac{1}{2}(e^x + e^{-x}) \end{pmatrix} \qquad 2. \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix}$$

Question 4. Verify whether the given equalities hold for the following matrices.

$$A = \begin{pmatrix} 2 & -3 \\ 4 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

1. $(A^\top)^{-1} = (A^{-1})^\top$
2. $(A^{-1})^{-1} = A$
3. $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
4. $(ABC)^\top = C^\top B^\top A^\top$

Question 5. Find the matrix A with the given information

1. $(7A)^{-1} = \begin{pmatrix} -3 & 7 \\ 1 & -2 \end{pmatrix}$
2. $(5A^\top)^{-1} = \begin{pmatrix} -3 & -1 \\ 5 & 2 \end{pmatrix}$
3. $(I + 2A)^{-1} = \begin{pmatrix} -1 & 2 \\ 4 & 5 \end{pmatrix}$
4. $A^{-1} = \begin{pmatrix} 2 & -1 \\ 3 & 5 \end{pmatrix}$

For each matrix A , compute:

- (a) A^3
- (b) A^{-3}
- (c) $A^2 - 2A + I$

Question 6. Compute $p(A)$ for the given matrix A and the following polynomials.

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 0 \\ 4 & 1 \end{pmatrix}$$

- (a) $p(x) = x - 2$
- (b) $p(x) = 2x^2 - x + 1$
- (c) $p(x) = x^3 - 2x + 1$

Question 7. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Question 8. If a polynomial $p(x)$ can be factored as a product of lower degree polynomials, say

$$p(x) = p_1(x)p_2(x)$$

and if A is a square matrix, then it can be proved that

$$p(A) = p_1(A)p_2(A).$$

Let

$$p(x) = x^2 - 9, \quad p_1(x) = x + 3, \quad p_2(x) = x - 3$$

1. Verify this statement for the above polynomials and the following matrices

$$A_1 = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 0 \\ 4 & 1 \end{pmatrix}$$

2. Prove that for any square matrix A , $p(A) = p_1(A)p_2(A)$

Question 9.

1. Give an example of two 2×2 matrices such that

$$(A + B)(A - B) \neq A^2 - B^2$$

2. Find a valid formula for multiplying out $(A + B)(A - B)$.
3. Establish the conditions on matrices A and B , under which the equality

$$(A + B)(A - B) = A^2 - B^2$$

is valid.

4. Find a valid formula for $(A + B)^3$.
5. Find a valid formula for $(A - B)^3$.

Question 10. Show that if a square matrix A satisfies the equation

$$A^2 + 2A + I = 0$$

then A must be invertible. Find the inverse of A in this case.

Question 11. Show that if $p(x)$ is a polynomial with a nonzero constant term, and if A is a square matrix for which $p(A) = 0$, then A is invertible.

Question 12. Is it possible for $A^3 = I$ without A being invertible? Why?

Question 13. Can a matrix with a row of zeros or a column of zeros have an inverse? Why?

Question 14. Simplify the given expression assuming that A, B, C, D are invertible.

1. $(AB)^{-1}(AC^{-1})(D^{-1}C^{-1})^{-1}D^{-1}$
2. $(AC^{-1})^{-1}(AC^{-1})(AC^{-1})^{-1}AD^{-1}$

Question 15. Assuming that all matrices are $n \times n$ and invertible, solve for D

1. $C^T B^{-1} A^2 B A C^{-1} D A^{-2} B^T C^{-2} = C^T$
2. $ABC^T DBA^T C = AB^T$

Question 16. Determine whether the given matrix is elementary

1. $\begin{pmatrix} 1 & 0 \\ -5 & 1 \end{pmatrix}$
2. $\begin{pmatrix} -5 & 1 \\ 1 & 0 \end{pmatrix}$
3. $\begin{pmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{pmatrix}$
4. $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
5. $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$
6. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 9 \\ 0 & 0 & 1 \end{pmatrix}$
7. $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$
8. $\begin{pmatrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$
9. $\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Question 17. Find a row operation and the corresponding elementary matrix that will transform the given elementary matrix to the identity matrix.

$$1. \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$$

$$2. \begin{pmatrix} -7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$3. \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{pmatrix}$$

$$4. \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$5. \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$$

$$6. \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$7. \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$8. \begin{pmatrix} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Question 18. In the following examples, an elementary matrix E and a matrix A are given. Identify the row operation corresponding to E and verify that the product EA results from applying the row operation to A .

$$1. E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{pmatrix}$$

$$2. E = \begin{pmatrix} -6 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{pmatrix}$$

$$3. E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{pmatrix}$$

$$4. E = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{pmatrix}$$

$$5. E = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$$6. E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Question 19. Use the following matrices and find an elementary matrix E that satisfies the

stated equation.

$$A = \begin{pmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{pmatrix},$$

$$C = \begin{pmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} 8 & 1 & 5 \\ -6 & 21 & 3 \\ 3 & 4 & 1 \end{pmatrix},$$

$$F = \begin{pmatrix} 8 & 1 & 5 \\ 8 & 1 & 1 \\ 3 & 4 & 1 \end{pmatrix}$$

- | | |
|-------------|-------------|
| 1. $EA = B$ | 2. $EA = C$ |
| 3. $EB = A$ | 4. $EC = A$ |
| 5. $EB = D$ | 6. $ED = B$ |
| 7. $EB = F$ | 8. $EF = B$ |

Question 20. Determine the inverse matrix A^{-1} (if the inverse exists) for each of the given matrices using the matrix inversion algorithm. Identify the sequence of elementary matrices $E_1, E_2, \dots$ associated with each row operation performed throughout the process.

1. $A = \begin{pmatrix} 1 & 4 \\ 2 & 7 \end{pmatrix}$

2. $A = \begin{pmatrix} 2 & -4 \\ -4 & 8 \end{pmatrix}$

3. $A = \begin{pmatrix} 1 & -5 \\ 3 & -16 \end{pmatrix}$

4. $A = \begin{pmatrix} 6 & 4 \\ -3 & -1 \end{pmatrix}$

5. $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$

6. $A = \begin{pmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{10} \\ \frac{1}{5} & -\frac{4}{5} & \frac{1}{10} \end{pmatrix}$

7. $A = \begin{pmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & 9 \end{pmatrix}$

8. $A = \begin{pmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} \\ \frac{2}{5} & -\frac{3}{5} & -\frac{3}{10} \\ \frac{1}{5} & -\frac{4}{5} & \frac{1}{10} \end{pmatrix}$

Question 21. Use the inversion algorithm to find the inverse of the matrix (if the inverse exists).

$$1. \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$2. \begin{pmatrix} \sqrt{2} & 3\sqrt{2} & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$3. \begin{pmatrix} 2 & 6 & 6 \\ 2 & 7 & 6 \\ 2 & 7 & 7 \end{pmatrix}$$

$$4. \begin{pmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{pmatrix}$$

$$5. \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{pmatrix}$$

$$6. \begin{pmatrix} 2 & -4 & 0 & 0 \\ 1 & 2 & 12 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & -4 & -5 \end{pmatrix}$$

$$7. \begin{pmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 2 & 1 & 5 & -3 \end{pmatrix}$$

$$8. \begin{pmatrix} 0 & 0 & 0 & 5 \\ 0 & 0 & -5 & 0 \\ 0 & -3 & 0 & 0 \\ 2 & 0 & 5 & -3 \end{pmatrix}$$

Question 22. Find the inverse of the given matrix, where $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α are all nonzero constants.

$$1. \begin{pmatrix} \alpha_1 & 0 & 0 & 0 \\ 0 & \alpha_2 & 0 & 0 \\ 0 & 0 & \alpha_3 & 0 \\ 0 & 0 & 0 & \alpha_4 \end{pmatrix}$$

$$2. \begin{pmatrix} \alpha & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \alpha & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$3. \begin{pmatrix} 0 & 0 & 0 & \alpha_1 \\ 0 & 0 & \alpha_2 & 0 \\ 0 & \alpha_3 & 0 & 0 \\ \alpha_4 & 0 & 0 & 0 \end{pmatrix}$$

$$4. \begin{pmatrix} \alpha & 0 & 0 & 0 \\ 1 & \alpha & 0 & 0 \\ 0 & 1 & \alpha & 0 \\ 0 & 0 & 1 & \alpha \end{pmatrix}$$

Question 23. Find all values of c , if any, for which the given matrix is invertible

$$1. A = \begin{pmatrix} c & c & c \\ 1 & c & c \\ 1 & 1 & c \end{pmatrix}$$

$$2. B = \begin{pmatrix} c & 1 & 0 \\ 1 & c & 1 \\ 0 & 1 & c \end{pmatrix}$$

Question 24. Express the given matrix A and its inverse as products of elementary matrices

$$1. A = \begin{pmatrix} -3 & 1 \\ 2 & 2 \end{pmatrix}$$

$$2. A = \begin{pmatrix} 1 & 0 \\ -5 & 2 \end{pmatrix}$$

$$3. A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$4. A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Question 25. For each of the following pairs of matrices A and B , show that the matrices A and B are row equivalent by

- (a) Find a sequence of elementary row operations that produce B from A .
- (b) Find a sequence of elementary row operations that produce A from B .
- (c) Use the results from 1 to find a matrix C such that $CA = B$.
- (d) Use the results from 2 to find a matrix D such that $DB = A$.

$$1. A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 1 \\ 2 & 1 & 9 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 2 & -2 \\ 1 & 1 & 4 \end{pmatrix}$$

$$2. A = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 1 & 0 \\ 3 & 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 6 & 9 & 4 \\ -5 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

Question 26. Show that

$$A = \begin{pmatrix} 0 & a & 0 & 0 & 0 \\ b & 0 & c & 0 & 0 \\ 0 & d & 0 & e & 0 \\ 0 & 0 & f & 0 & g \\ 0 & 0 & 0 & h & 0 \end{pmatrix}$$

is singular for any values of a, b, c, d, e, f, g, h .

Question 27. Suppose that each of the following is the augmented matrix for a linear system. Solve the systems by inverting the coefficient matrix (if the inverse exists) and apply the following theorem

Theorem 1 *Given $A \in \mathcal{M}_{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$, if A is invertible, then the system of equations $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.*

$$1. \left(\begin{array}{cc|c} 2 & 0 & -1 \\ 3 & 2 & 0 \end{array} \right)$$

$$3. \left(\begin{array}{ccc|c} 1 & 0 & 8 & 6 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

$$5. \left(\begin{array}{ccc|c} 1 & 0 & -2 & 3 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

$$7. \left(\begin{array}{cccc|c} 1 & 0 & 0 & 3 & -2 \\ 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 2 & 3 \end{array} \right)$$

$$9. \left(\begin{array}{ccc|c} 2 & 1 & -5 & -20 \\ 0 & 2 & 2 & 7 \\ 3 & 1 & -9 & -36 \end{array} \right)$$

$$2. \left(\begin{array}{cccc|c} 3 & 0 & 0 & 1 & -4 \\ 3 & 0 & 2 & 1 & 7 \\ -1 & 3 & 0 & -2 & 4 \\ 0 & 0 & -1 & 2 & 1 \end{array} \right)$$

$$4. \left(\begin{array}{ccc|c} 1 & 0 & -7 & 8 \\ 0 & 0 & 3 & 2 \\ 0 & 1 & 1 & -5 \end{array} \right)$$

$$6. \left(\begin{array}{ccc|c} 5 & 20 & -18 & -11 \\ 3 & 12 & -14 & 3 \\ -4 & -16 & 13 & 13 \end{array} \right)$$

$$8. \left(\begin{array}{ccc|c} -2 & 1 & 1 & 15 \\ 6 & -1 & -2 & -36 \\ 1 & -1 & -1 & -11 \end{array} \right)$$

For Questions 28 – 30 do the following

Solve the given linear system by inverting the coefficient matrix (if it is invertible) and applying Theorem 1.

Question 28.

1.

$$\begin{aligned}x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10\end{aligned}$$

2.

$$\begin{aligned}2x_1 + 2x_2 + 2x_3 &= 0 \\ -2x_1 + 5x_2 + 2x_3 &= 1 \\ 8x_1 + x_2 + 4x_3 &= -1\end{aligned}$$

3.

$$\begin{aligned}x - y + 2z - w &= -1 \\ 2x + y - 2z - 2w &= -2 \\ -x + 2y - 4z - 2w &= 1 \\ 3x - 3w &= -3\end{aligned}$$

4.

$$\begin{aligned}-2y + 3z &= 1 \\ 3x + 6y - 3z &= -2 \\ 6x + 6y + 3z &= 5\end{aligned}$$

5.

$$\begin{aligned}5x_1 - 5x_2 - 15x_3 &= 40 \\ 4x_1 - 2x_2 - 6x_3 &= 19 \\ 3x_1 - 6x_2 - 17x_3 &= 41\end{aligned}$$

6.

$$\begin{aligned}x_1 + 3x_2 - x_3 &= 0 \\ x_2 - 8x_3 &= 0 \\ 4x_3 &= 0\end{aligned}$$

7.

$$\begin{aligned}2x_1 + x_2 + 3x_3 &= 0 \\ x_1 + 2x_2 &= 0 \\ x_2 + x_3 &= 0\end{aligned}$$

8.

$$\begin{aligned}2x - y - 3z &= 0 \\ -x + 2y - 3z &= 0 \\ x + y + 4z &= 0\end{aligned}$$

Question 29.

1.

$$\begin{aligned} -2x_1 + x_2 + 8x_3 &= 0 \\ 7x_1 - 2x_2 - 22x_3 &= 0 \\ 3x_1 - x_2 - 10x_3 &= 0 \end{aligned}$$

2.

$$\begin{aligned} 5x_1 - 2x_3 &= 0 \\ -15x_1 - 16x_2 - 9x_3 &= 0 \\ 10x_1 + 12x_2 + 7x_3 &= 0 \end{aligned}$$

3.

$$\begin{aligned} 2x_1 - 3x_2 + 4x_3 - x_4 &= 0 \\ 7x_1 + x_2 - 8x_3 + 9x_4 &= 0 \\ 2x_1 + 8x_2 + x_3 - x_4 &= 0 \end{aligned}$$

4.

$$\begin{aligned} v + 3w - 2x &= 0 \\ 2u + v - 4w + 3x &= 0 \\ 2u + 3v + 2w - x &= 0 \\ -4u - 3v + 5w - 4x &= 0 \end{aligned}$$

5.

$$\begin{aligned} 2x + 2y + 4z &= 0 \\ w - y - 3z &= 0 \\ 2w + 3x + y + z &= 0 \\ -2w + x + 3y - 2z &= 0 \end{aligned}$$

6.

$$\begin{aligned} 2x_1 + 6x_2 + 13x_3 + x_4 &= 0 \\ x_1 + 4x_2 + 10x_3 + x_4 &= 0 \\ 2x_1 + 8x_2 + 20x_3 + x_4 &= 0 \\ 3x_1 + 10x_2 + 21x_3 + 2x_4 &= 0 \end{aligned}$$

7.

$$\begin{aligned} 2x_1 - 6x_2 + 3x_3 - 21x_4 &= 0 \\ 4x_1 - 5x_2 + 2x_3 - 24x_4 &= 0 \\ -x_1 + 3x_2 - x_3 + 10x_4 &= 0 \\ -2x_1 + 3x_2 - x_3 + 13x_4 &= 0 \end{aligned}$$

Question 30.

1.

$$\begin{aligned}x + y &= 1 \\4x + 3y &= 2\end{aligned}$$

2.

$$\begin{aligned}-3x + y &= 1 \\4x + 2y &= 0\end{aligned}$$

3.

$$\begin{aligned}3x - 3y &= 3 \\4x - y - 3z &= 3 \\-2x - 2y &= -2\end{aligned}$$

4.

$$\begin{aligned}2x - 4z &= 1 \\4x + 3y - 2z &= 0 \\2x + 2z &= 0\end{aligned}$$

5.

$$\begin{aligned}x + 2y + z &= 1 \\2x + 3y + 2z &= 0 \\x + y + z &= 2\end{aligned}$$

6.

$$\begin{aligned}2x - 4z &= 1 \\4x + 3y - 2z &= 0 \\2x + 2z &= 2\end{aligned}$$

Question 31. Show that the matrices

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \quad B = \begin{pmatrix} d & e \\ 0 & f \end{pmatrix}$$

commute if and only if

$$\begin{vmatrix} b & a - c \\ e & d - f \end{vmatrix} = 0$$

Question 32. Show that the matrices

$$A = \begin{pmatrix} a & 0 \\ b & c \end{pmatrix}, \quad B = \begin{pmatrix} d & 0 \\ e & f \end{pmatrix}.$$

commute if and only if

$$\begin{vmatrix} b & a - c \\ e & d - f \end{vmatrix} = 0$$